SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

umhackathon@um.edu.my

© UMHackathon 2026

Email: umhackathon@um.edu.my

1. Introduction

1.1. Purpose

The system analysis document highlights the technical scope and design-related decisions behind the development of "TradeBridge ASEAN: Cross-Border SME Pricing & Market Entry Intelligence". The key elements of the system covered are:

1. Architecture: The system operates as an asynchronous, event-driven orchestration pipeline built natively on n8n.
2. Data Flows: The system synthesizes structured inventory data from Google Sheets with real-time unstructured market data via web searches to generate actionable financial decisions.
3. Model Process: The end-to-end workflow involves chronological daily scans, fetching external forex APIs, and triggering GLM-5.1 for reasoning, culminating in 1-by-1 Telegram messages for asynchronous human-in-the-loop (HITL) approval.
4. Role of Reference: This document serves as a high-level architectural reference for the development and QA teams to ensure successful deployment during the 10-day sprint.

1.2. Background

Malaysian micro, small, and medium enterprises (MSMEs) currently struggle to navigate fluctuating exchange rates, opaque cross-border logistics fees, varying import duties, and localized competitor pricing.

1. Previous Version: MSMEs traditionally rely on manual, error-prone spreadsheet calculations or static markups, which often lead to eroded profit margins or uncompetitive market pricing.
2. New Capabilities: TradeBridge ASEAN removes this friction by acting as a financial co-pilot. It automates the calculation of volatile variables (e.g., Thai VAT, Philippine de minimis thresholds) so merchants can confidently maintain baseline profitability.

© UMHackathon 2026

1.3. Target Stakeholder

Stakeholders	Roles	Expectations
SME Owner / Founder	Receive and review individual price adjustment recommendations.	To receive 1-by-1 Telegram messages with reasoned price adjustments for easy, calculation-free approval.
E-commerce Manager	Manage cross-border product listings and localized pricing.	Automated retrieval of regional tax data (e.g., Thai VAT) and accurate landed cost calculations.
Logistics Analyst	Monitor shipping costs against product value.	The AI should accurately suggest unlisting items if shipping costs exceed the product's value to prevent operating at a loss.
Development Team	Build and maintain the n8n orchestration and GLM-5.1 API integration.	Clear schema designs, stable webhook routing, and optimized prompt configurations for the LLM.
QA Team	Validate the system behavior, integration testing, and edge cases.	Ensure adversarial AI testing passes and webhook timeouts/failures are handled gracefully.

© UMHackathon 2026

2.0 System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview

Type	Details
System	Conversational Command Interface (Telegram Bot) & Database (Google Sheets).
Architecture	Asynchronous, event-driven orchestration pipeline (n8n).

TradeBridge ASEAN operates primarily as a high-speed generative reasoning engine. The frontend user interface is handled entirely via a conversational Telegram Bot integration. The backend consists of an n8n orchestration server that communicates with Google Sheets for MVP data persistence and fetches real-time data via external Forex APIs.

2.1.2. LLM as Service Layer

The Z.AI GLM-5.1 model serves as the exclusive cognitive reasoning engine. It leverages a 200,000-token context window and native web search tool integration to process long-horizon agentic workflows.

2.1.3. Dependency Diagram Flow

a. Prompts are constructed using a zero-shot agentic prompting strategy, explicitly instructed to prioritize stability and only propose changes if margins fall below strict thresholds. **b.** The context window receives semantic batches (via n8n's "Split in Batches" node) alongside static system instructions and repetitive categorical web search results, utilizing Z.AI's Context Caching to reduce token consumption. **c.** If GLM-5.1 breaks the requested JSON schema or hallucinates, the n8n structured output parser automatically routes the response to a failure branch for a single retry. **d.** To enforce API limits, the n8n workflow utilizes pacing delays to

© UMHackathon 2026

prevent exhaustion of the Google Sheets API strict 60 requests per minute quota. **e.** On-demand updates trigger the n8n webhook via a /run_update command in Telegram, initiating a targeted inventory scan.

2.2. Technological Stack

- 2.2.1. Frontend: Telegram Bot (Conversational Command Interface).
- 2.2.2. Backend (Orchestration): n8n (Handles cron jobs, Split-in-Batches loops, and webhooks).
- 2.2.3. Database: Google Sheets API (Bidirectional MVP data persistence layer).
- 2.2.4. AI / Engine: Z.AI GLM-5.1 (Cognitive reasoning engine & native web search).

2.3. Key Data Flows

2.3.1. Data Flow Diagram (DFD) Journey

1. **Daily Scan:** The n8n cron job fetches the Google Sheets inventory chronologically, followed by external forex APIs.
2. **Inference:** n8n triggers GLM-5.1 for web searches and reasoning to evaluate if a price change is mathematically required.
3. **HITL Approval:** If a change is proposed, n8n sends a Telegram message with interactive inline buttons, placing the execution thread into a "Wait for Webhook" state.
4. **Database Write:** Once the user taps "Approve", the webhook resumes the workflow and n8n writes the new price back to Google Sheets.

© UMHackathon 2026

3. Functional Requirements & Scope

3.1. Minimum Viable Product

No.	Features	Description
1	Intelligent Market Scanning	Daily automated retrieval of localized logistics costs, import duties, and competitor pricing.
2	Sequential HITL Approvals	Individual price change recommendations sent via Telegram with interactive approve/reject capabilities.
3	Strategic Market Delisting	Autonomous capability to suggest unlisting a product in a specific country if shipping costs mathematically negate profitability.
4	Conversational Command Interface	Telegram bot integration allowing users to trigger workflows manually and list items dynamically.

© UMHackathon 2026

3.2. Non-Functional Requirements (NFRs)

Quality	Requirements	Implementation
Cost Efficiency	Must manage the high inference costs associated with evaluating 500 items daily across ASEAN at $1.40/M input tokens.	Utilizes semantic batching via n8n and Z.AI Context Caching for repetitive search results, reducing token consumption by over 80%.
Reliability	Must handle LLM hallucinations and broken JSON schemas without corrupting database integrity.	The n8n structured output parser routes failures to a retry branch; if it fails again, it defaults to a safe state (skipping the item) and notifies the user.
Performance Constraints	The system must not exceed API quota thresholds and prevent server memory deadlocks.	Implements pacing delays for Google Sheets (60 req/min limit) and an absolute 12-hour webhook timeout where unapproved changes gracefully default to rejection.

© UMHackathon 2026

3.3. Out of Scope / Future Enhancements

- Direct e-commerce API integration (e.g., syncing directly to live Shopee/Lazada storefronts).
- Vision processing capabilities (e.g., creating items from uploaded photos).
- Production-grade backend relational databases (e.g., PostgreSQL).
- Multilingual NLP parsing for user commands (Telegram UI remains strictly English).

4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation

- 4.1.1. Fallback & Failure Rollback: If the system fails to parse the GLM-5.1 output correctly after a retry, the workflow applies an emergency fallback to preserve the legacy price and skips the item modification.

4.2. Monitoring

- **4.2.1.** The n8n environment actively monitors the "Wait for Webhook" execution states. Any webhook pending for over 12 hours triggers an automatic cancellation and rejection state to keep the orchestration pipeline healthy.

4.3. Assumptions

- **4.3.1.** The system heavily relies on the assumption that the primary key ("Item Name") matches exactly between the Telegram commands and the Google Sheets rows.

© UMHackathon 2026

4.4. External Dependencies

Tools	Purpose	Risks
Z.AI GLM-5.1 API	Acts as the core cognitive reasoning engine for prompt inference, multi-step reasoning, and web scraping.	High risk if web search cannot bypass e-commerce anti-bot countermeasures to accurately scrape competitor prices.
n8n Orchestrator	Handles workflow loops, cron jobs, semantic batching, and webhook suspensions.	Medium risk if master daily execution latency exceeds time limits imposed by standard hosting environments.
Google Sheets API	Serves as the MVP bidirectional database persistence layer.	High risk of data synchronization failures if write requests exceed 60 per minute or if users bypass the bot to create manual duplicates.

5. Project Management & Team Contributions

5.1. Project Timeline

The duration of the Preliminary round is approximately 10 days considering the sprint structure.

- Days 1-2: n8n deployment and Database schema design.
- Days 4-5: Google Sheets connection, Telegram bot listener setup, and Core workflow loops.

© UMHackathon 2026

- Days 6-7: GLM-5.1 integration, Interactive UI creation, and JSON schema enforcement.
- Days 8-10: Orchestration stabilization, full integration testing, and final documentation.

5.2. Team Members Role

Team Member	Role	Key Responsibilities	Deliverables (10-Day Sprint)	Dependencies
Gan Ching Kang	Backend / Orchestration Lead	n8n server setup, cron jobs, Split-in-Batches loops, workflow orchestration	• -n8n deployed (Day 2) • - Core workflow loops (Day 5) • Stable orchestration pipeline (Day 8)	Depends on DB schema (Tan Weiye), AI endpoints (Chin Jia Den)
Chin Jia Den	AI Engineer	GLM-5.1 API integration, prompt engineering, JSON schema enforcement, caching	• GLM API integration (Day 6) • Structured JSON outputs (Day 7) • Prompt optimization + caching (Day 8)	Needs orchestration hooks (Gan), DB structure (Tan)
Tai Jia Xuan	Frontend / UI Developer	TelegramBot integration, inline keyboards, command parsing	• Telegram bot listener (Day 4) • Interactive approval UI (Day 6) • Command parsing stable (Day 8)	Needs backend webhook endpoints (Gan)
Tan Weiye	Database Administrator	Google Sheets API, schema design, Forex API integration	• Schema design (Day 2) • Google Sheets connected (Day 4) • Forex API integrated (Day 6)	Foundation for backend + AI
Yek Zu Xian	QA Engineer	Integration testing, webhook edge cases, adversarial AI testing, documentation (PRD, SAD, QATD)	• Test cases (Day 5) • Full integration testing (Day 8–9) • Final documentation (Day 10)	Requires all modules

© UMHackathon 2026